

Effect analysis of oxide layer on thermal hydraulic characteristics of LBE cooled reactor

Chenglong Wang^{*}, Chen Wang, Wenxi Tian, Guanghui Su, Suizheng Qiu

Department of Nuclear Science and Technology, State Key Lab. of Multiphase Flow in Power Engineering, Shaanxi Key Lab. of Advanced Nuclear Energy and Technology, Xi'an Jiaotong University, Xi'an, Shaanxi 710049, China

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ABSTRACT

For Lead-Bismuth Eutectic (LBE) alloy cooled reactor, oxidation corrosion is very important and would affect the thermal hydraulic characteristics of the reactor. In this paper, based on LETHAC code developed by Xi'an Jiaotong University (XJTU) for system level thermal hydraulic analysis LBE reactor, a simulation module for oxidation corrosion is developed and added. The influence of oxidation layer on the heat transfer characteristics of the reactor is conducted. Based on LESMOR reactor designed by XJTU, a simulation calculation is carried out under the long-term operation conditions of the reactor considering the growth of oxidation layer. The distribution of the oxide layer in the reactor, the change of temperature appreciation caused by the oxide layer over time and the distribution in the reactor are obtained. The presence of oxide layer will affect the fluid temperature and wall temperature of reactor. The longer the run time, the greater the impact. This paper could provide solid analysis methodology reference for the design of LBE reactor.

1. Introduction

As one of the fourth-generation nuclear reactor, lead cooled fast reactor has multiple advantages, such as lead based alloy stable chemical properties, high operating temperatures, large natural circulation capacity, good inherent safety and economy (Ujp, 2002). Lead cooled fast reactor uses lead-bismuth Eutectic (LBE) as coolant of primary side. LBE is a liquid metal with stable chemical properties, but it will corrode structural steel. With the corrosion of the structural steel, the flow of LBE would be blocked due to the gradual materials precipitation in the non-isothermal flow process.(Zhang, 2009).

A protective oxide layer could be formed on the surface of structural steel by injecting a certain concentration of oxygen into LBE coolant (Zhang and Li, 2008). With the diffusion and accumulation of oxygen in the reactor system, oxide layers will accumulate on the surface of structural materials such as pipe wall and fuel rods. When oxide layers accumulate to a given amount on the wall surface, flow and heat transfer performance of the reactor system will be affected and additional thermal resistance will be created on the wall surface which could damage the reactor safety.

In the open literature, much works has been done on the accumulation of oxide layer in lead-cooled fast reactor. The experimental

facility DELTA has been built in Los Alamos National Laboratory(LANL) to evaluate the fouling generated by the oxide layer of the heat exchanger under different oxygen concentrations(Ilnce, 2006). An experimental facility named CORRIDA was built by Karlsruhe Institute of Technology (KIT) to evaluate the diffusion of oxygen in the LBE and the accumulation of oxide layer in the facility(Schroer, 2010). COLONRI facility was built by NRI, to evaluate the interaction of liquid LBE and structural materials(Barbier et al., 2001; Zhang et al., 2005). There is a few investigations using numerical simulation method. A simulation using OpenFoam was conducted by Yan et al. to evaluate the oxidation corrosion phenomena in the inner cylinder section of DELTA bench. Finally, the distribution of oxide layer in DELTA bench was achieved.

However, most of the studies are long-term research based on experimental facilities, and main research contents of these studies are focused on the accumulation and corrosion of oxide layer itself. The experimental research results on the oxidation corrosion of LBE have been summarized to form some empirical/semi empirical relationships for predicting the formation of oxide layers in LBE oxidation corrosion. Thermal-hydraulic analysis code is an effective method to evaluate the thermal-hydraulic characteristics of the reactor system. But traditional thermal-hydraulic analysis codes ignore the strong corrosion effect of LBE fluid. These codes cannot calculate the accumulation and corrosion process of oxide layer for the reactor system, nor can they evaluate the

^{*} Corresponding author.

E-mail address: chlwang@mail.xjtu.edu.cn (C. Wang).

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Nomenclature			
C_O	Concentration of oxygen, wt%	d	Hydraulic diameter
C_{Fe}	Concentration of iron, wt%	K_P	Oxide layer growth factor
U	Circumference of the pipe section, m	ρ	Density
A	Area of flow, m^2	λ	Thermal conductivity
j_O	Oxygen flux, $m\cdot wt\%/s$	T	Temperature
j_{ox}	Oxide flux, $m\cdot wt\%/s$	Nu	Nusselt number
$\dot{q}$	Oxygen source, $wt\%/s$	Re	Reynold number
j_{Fe}	Iron flux $m\cdot wt\%/s$	Pe	Peclet number
f_O^{Fe}	Proportion of oxygen in Fe_3O_4	η	Rough surface correction factor
f_{ox}^{me}	Proportion of oxygen in the oxide layer	η_o	Oxide layer correction factor
M_O	Molar mass of oxygen	q	Heat flux
M_{Fe}	Molar mass of iron	<i>rough</i>	Rough surface
δ	Thickness of oxide layer	<i>smooth</i>	Smooth surface
		<i>mag</i>	magnetite
		<i>sp</i>	Spinel

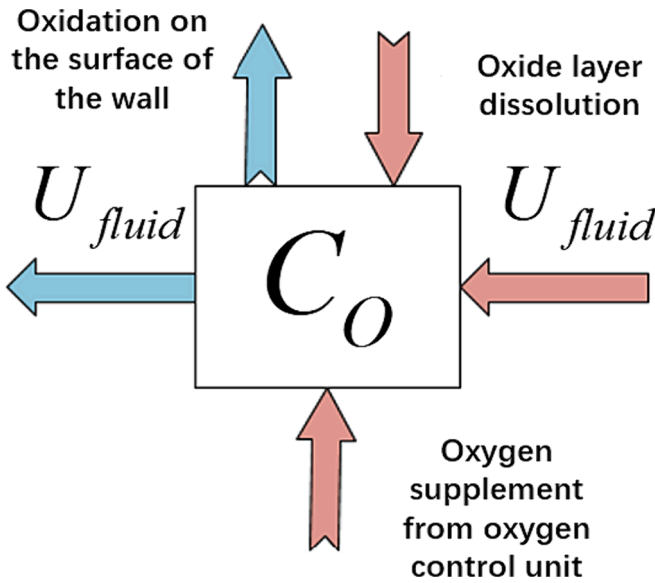


Fig. 1. Generation and consumption of oxygen.

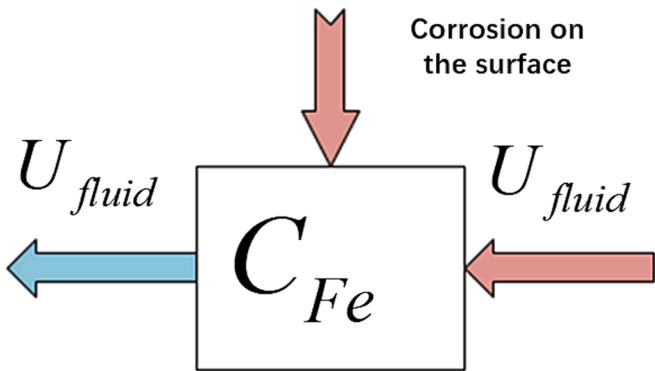


Fig. 2. Generation and consumption of iron.

long-term impact of oxide layer on the thermal hydraulic characteristics of the reactor. Besides, the effect of oxide layer on the flow and heat transfer performance has not been investigated. Therefore, it is necessary to develop a code combining oxidation corrosion calculation and thermal hydraulic calculation. It aims to evaluate changes in

thermal-hydraulic characteristics due to oxide layer accumulation during long-term reactor operation.

In the present paper, based on the lead-cooled fast reactor thermal-hydraulic transient analysis code LETHAC (Wei et al., 2021), an oxidation corrosion analysis module is developed. Then, a set of coupling calculations are conducted for the LESMOR reactor based on the LETHAC thermal-hydraulic calculation and oxidation corrosion calculation modules. Based on the calculation result, the time-dependent changes of the reactor steady-state thermal hydraulic characteristics considering oxidation corrosion are evaluated.

2. Numerical method

2.1. Accumulation and corrosion model of oxide layer

2.1.1. Diffusion of oxygen and iron in LBE fluid

In the LBE reactor system, the concentration of oxygen varies as the LBE fluid circulates through the circuit in the system. When the fluid flows through the oxygen control unit (liquid or solid oxygen control tank), oxygen concentration in the fluid rises. When the fluid contacts with steel walls, it corrodes the wall surface, then the oxide layer or the steel matrix dissolved. The variation of oxygen concentration in LBE fluid is shown in Fig. 1.

When the steel matrix of the walls is corroded, iron concentration will rise as Fig. 2:

Based on the introduction above, a set of governing equations of oxygen and iron in LBE fluid can be obtained as Eq. (1) and Eq. (2):

$$\frac{\partial C_O}{\partial t} + u \frac{\partial C_O}{\partial x} = \frac{U}{A} j_O + \dot{q} \quad (1)$$

$$\frac{\partial C_{Fe}}{\partial t} + u \frac{\partial C_{Fe}}{\partial x} = \frac{U}{A} j_{Fe} \quad (2)$$

In Eq. (1), transport flux of oxygen j_O is as Eq. (3):

$$j_O = -f_{ox}^{me} j_{ox} - f_O^{Fe} j_{Fe} \quad (3)$$

Assuming that iron and oxygen in the LBE fluid are all combined to form Fe_3O_4 , then the factor f_O^{Fe} is as Eq. (4):

$$f_O^{Fe} = \frac{4M_O}{3M_{Fe}} \quad (4)$$

In the experimental studies, it has been known that the main composition of the oxide layer for T91 steel is a double-layer structure of magnetite and iron-chromium spinel ($Fe_{1-x}Cr_xO_4$). Therefore, the factor f_{ox}^{me} is as Eq. (5):

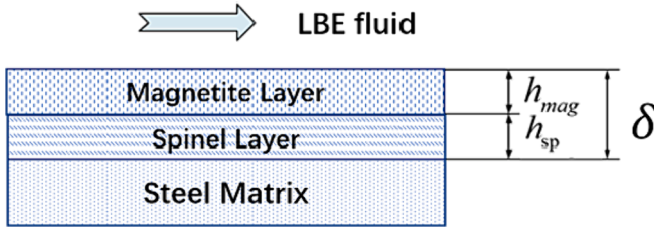


Fig. 3. Structure of double layer oxide layer.

$$f_{ox}^{me} = \frac{1}{2} \left(\frac{4M_O}{4M_O + \alpha M_{Fe} + \beta M_{Cr}} + \frac{4M_O}{4M_O + 3M_{Fe}} \right) \quad (5)$$

2.1.2. Accumulation and corrosion of oxide layer

The generation of the oxide layer requires a minimum oxygen concentration $C_{O,min}$, when the current oxygen concentration is higher than minimum oxygen concentration, the oxide layer starts to accumulate, otherwise the oxide layer or the steel matrix starts being dissolved. Assuming that Fe_3O_4 is generated when the oxide layer is generated, then the minimum oxygen concentration required for oxide layer accumulation can be deduced according to the minimum Gibbs free energy of Fe_3O_4 generation. The minimum oxygen concentration is as Eq. (6). (Alm,afb-c, aat, 2008)

$$\log C_{O,min} = 7.858 - \frac{21840.25}{T} - \frac{3}{4} \log C_{Fe} \quad (6)$$

When fluid oxygen concentration is higher than minimum oxygen concentration, the accumulation and dissolution of oxide layer occur at the same time. The growth rate of oxide layer is as Eq. (7): (Martinelli et al., 2008)

$$\Delta \delta = \left(\frac{K_P}{\delta} - \frac{j_{Fe} \rho_{LBE}}{f_{ox}^{Fe} \rho_{ox}} \right) t \quad (7)$$

When fluid oxygen concentration is lower than minimum oxygen concentration, oxide layer starts being dissolved. The dissolution rate of oxide layer is as Eq. (8) (Martinelli et al., 2008)

$$\Delta \delta_{oxide} = - \frac{j_{Fe} \rho_{LBE}}{f_{ox}^{Fe} \rho_{ox}} \quad (8)$$

When the oxide layer is completely dissolved, i.e., $\delta_{oxide} \leq 0$, the steel matrix starts being dissolved. The dissolution rate is as Eq. (9): (Feng et al., 2021)

$$\Delta \delta_{matrix} = - \frac{j_{Fe} \rho_{LBE}}{f_{ss}^{Fe} \rho_{ss}} \quad (9)$$

2.2. Effect of oxide layer on heat transfer performance

2.2.1. Experimental model

When the oxide layers accumulate on the pipe wall, an additional heat transfer resistance will be formed on the surface of the pipe wall. The temperature difference between mainstream and wall will rise due to the present of this additional heat transfer resistance. The additional temperature difference is as Eq. (10).

$$T_w - T = \frac{q}{2\pi\lambda_{oxide}} \ln \left(\frac{r - \delta}{r} \right) \quad (10)$$

The thermal conductivity of oxide layer λ_{oxide} is determined by its composition. Due to the lack of parameters such as the porosity of the oxide layer structure, the porosity of the oxide layer structure is not considered for the time being. The thermal conductivity of Fe_3O_4 and Cr_2O_3 , the main components of the oxide layer, are shown in Eq. (11) and Eq. (12) (Saito et al., 1985).

$$\lambda_{Fe_3O_4} = 0.0423 - 1.37 \times 10^{-5} T \quad (11)$$

$$\lambda_{Cr_2O_3} = \frac{1}{0.037 + 1.6 \times 10^{-4} T} \quad (12)$$

The present of the oxide layer will also make the pipe surface rough. It affects the heat transfer performance of on the pipe wall. By using the lead–bismuth flow heat transfer experiment and the CFD method of different rough surfaces, the heat transfer relationship of different rough surfaces can be obtained as an Eq. (13). Its application range is $500 \leq Pe \leq 5000$.

$$Nu_{rough} = (6.55049 + 0.00612Pe^{0.88816})(0.99079 + 0.00559Pe^{0.71666} \left(\frac{\delta}{d} \right)^{0.48967}) \quad (13)$$

In the Eq. (13), the influence of rough surface on heat transfer performance appears in the form of correction factor η_r , as Eq. (14) shows.

$$Nu_{rough} = Nu_{smooth} \cdot \eta_r(Pe, \delta, d) \quad (14)$$

Therefore, the correction factor is as Eq. (15) shows:

$$\eta_r = \frac{Nu_{rough}}{Nu_{smooth}} = 0.99079 + 0.00559Pe^{0.717} \left(\frac{\delta}{d} \right)^{0.48967} \quad (15)$$

2.2.2. Theoretical model

For LBE fluid, a typical double layer oxide structure is shown in Fig. 3. A double layer oxide structure consists of an outer magnetite layer and an inner spinel layer. When the oxide layer accumulates on the pipe wall, the thermal resistance of the oxide layer itself becomes an additional thermal resistance of the pipe.

Therefore, considering the existence of oxide layer thermal resistance, the Nusselt number representing the pipeline can be formed as Eq. (16) by combining the original boundary before corrosion with the relationship between oxide layer thickness and wall thermal conductivity resistance. The effect oxide layer on Nusselt number can also be expressed in the form of correction factor η_o as Eq. (17).

$$Nu_{oxide} = Nu \frac{(T_w - T_b)}{(T_w - T_b) + q \left(\frac{\delta_{oxide}}{\lambda_{oxide}} - \frac{\delta_{r91}}{\lambda_{r91}} \right)} \quad (16)$$

$$\eta_o = \frac{T_w - T_b}{(T_w - T_b) + q \left(\frac{\delta_{oxide}}{\lambda_{oxide}} - \frac{\delta_{r91}}{\lambda_{r91}} \right)} \quad (17)$$

2.3. Thermal hydraulic models

2.3.1. Basic thermal hydraulic models

The thermal hydraulic models are integrated in the LETHAC code developed by Xi'an Jiaotong University (XJTU). LETHAC is a transient system level analysis code especially for LFR, therefore the thermal hydraulic models are established based on the properties of LBE and features of general components of LFR. In the LFR, liquid lead and lead–bismuth eutectic are two common coolants in the primary loop, these are fluids with high boiling points (1798 K for liquid lead and 1943 K for LBE). Therefore, the phenomenon of lead–bismuth two-phase boiling is usually not considered in the operation of LFR. The single-phase conservation equations are used in the code for lead-based coolant, as shown in Eqs. (18)–(20).

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial z} \left(\frac{W}{A} \right) = 0 \quad (18)$$

$$\frac{\partial}{\partial t} \left(\frac{W}{A} \right) + \frac{\partial}{\partial z} \left(\frac{W^2}{A^2} \right) = - \frac{\partial p}{\partial z} - \frac{fW|W|}{2D_e \rho A^2} - \rho g \quad (19)$$

$$\rho \frac{\partial h}{\partial t} + \frac{W}{A} \frac{\partial h}{\partial z} = \frac{qU}{A} \quad (20)$$

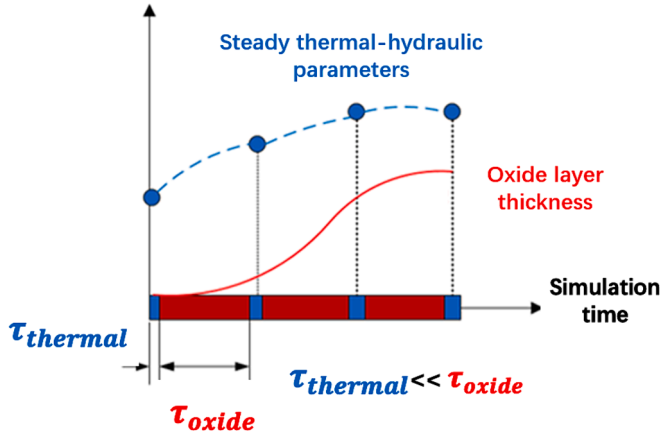


Fig. 4. Calculation results of coupled calculation module.

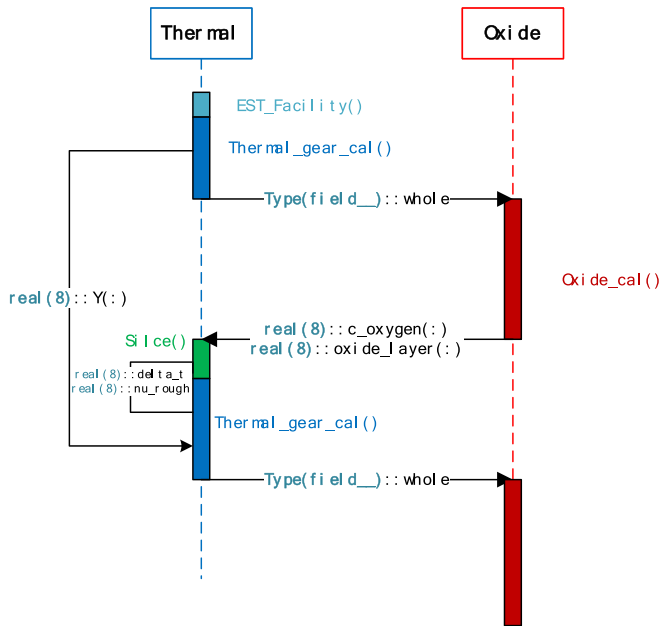


Fig. 5. Coupling process of thermal-hydraulic module and oxidation corrosion module.

In the secondary side of the heat exchanger or steam generator, two-phase water might present. Homogeneous flow model is used for two-phase water in the secondary side. In homogeneous model, the governing equations is similar to single-phase governing equations, but the coolant density is as Eq. (21).

$$\rho = \alpha \rho_v + (1 - \alpha) \rho_l \quad (21)$$

2.3.2. Flow and heat transfer models

In the present code, various of flow and heat transfer models are included. For heat transfer cases in different geometry, different heat transfer correlations are used in the present code. For heat transfer case in hexagon fuel bundles in the reactor core, the Ushakov correlation and Kazimi-Carelli correlations are used (Steiner et al., 2008).

When $P/D > 1.3$, Ushakov correlation is used:

$$Nu = 7.55 \left(\frac{P}{D} \right) - 20 \left(\frac{P}{D} \right)^{-13} + 3.67 \left(\frac{P}{D} \right) Pe^{0.56+0.19/(P/D)} \quad (22)$$

When $1.2 < P/D < 1.3$, Kazimi-Carelli correlation is used:

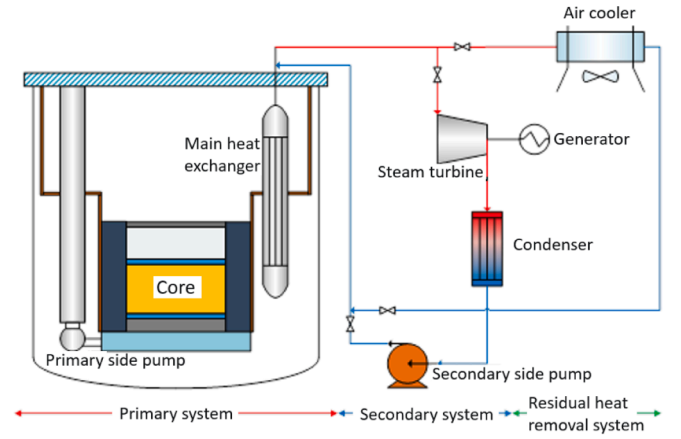


Fig. 6(a) System layout of LESMOR reactor

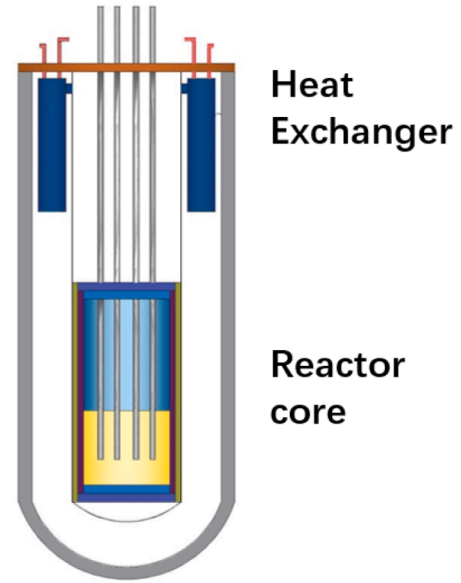


Fig. 6(b) Schematic of LESMOR reactor core

Fig. 6. Schematic of LESMOR reactor.

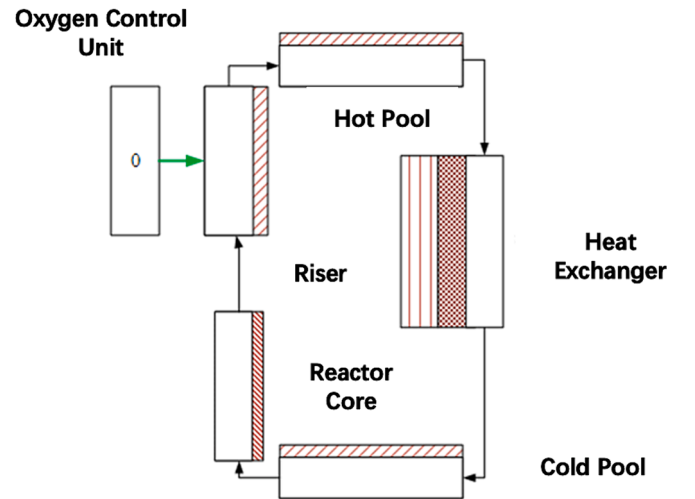


Fig. 7. Nodalized model of LESMOR reactor.

Table 1
Simulation parameters.

Simulation parameters	Value
Heating power	50 MW
Input temperature	370°C
Output temperature	400°C
Input temperature of secondary side	105°C
Initial oxidation concentration	$1.2 \times 10^{-6}\text{wt\%}$, $1.2 \times 10^{-5}\text{wt\%}$, $1.2 \times 10^{-4}\text{wt\%}$
Wall roughness	0, 10 μm , 100 μm
Mass of flow of secondary side	10 kg/s
Thermal hydraulic calculation time	100 s
Oxide corruption calculation time	5000 h

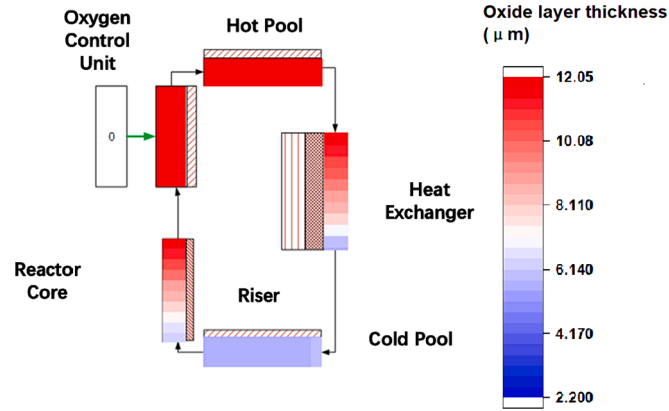


Fig. 8a. Distribution of oxide layer in the reactor.

$$Nu = 4.0 + 0.16 \left(\frac{P}{D} \right)^{5.0} + 0.33 \left(\frac{P}{D} \right)^{3.8} \left(\frac{Pe}{100} \right)^{0.86} \quad (23)$$

In circular tube, Cheng-Tak correlation is used: (Hwang and Lim, 2010)

$$Nu = A + 0.018Pe^{0.8} \quad (24)$$

$$A = \begin{cases} 4.5 & Pe \leq 1000 \\ 5.4 - 5.4 - 9 \times 10^{-4}Pe & 1000 \leq Pe \leq 2000 \\ 3.6 & Pe \geq 2000 \end{cases} \quad (25)$$

When calculating friction resistance coefficient of the hexagon bundle, Novendstern correlation is used. (Novendst.Eh., 1972)

When $Re < 400$, laminar friction resistance coefficient is as follows:

$$f_L = \frac{64}{Re} \quad (26)$$

When $Re > 5000$, turbulent friction resistance coefficient is as follows:

$$f_T = M \frac{0.3164}{Re^{0.25}} \quad (27)$$

When $400 < Re < 5000$:

$$M = \left[\frac{1.034}{\chi^{0.124}} + \frac{29.7\chi^{694}Re^{0.086}}{\left(\frac{H}{d_r} \right)^{2239}} \right]^{0.885} \quad (28)$$

$$f_{ir} = f_L(1 - \psi)^{0.5} + f_T\psi^{0.5} \quad (29)$$

$$\psi = \frac{Re - 400}{4600} \quad (30)$$

2.3.3. Properties of liquid and solid materials

In 2016, OECD/NEA published a handbook on lead/LBE. In this handbook, a set of properties of liquid lead and liquid LBE are summarized. (Oecd-nea., 2007) These properties models have high accuracy and certain application range to meet the requirement of operating requirements of LFR. In the present code, this set of property models are used to calculate the property of LBE in the primary loop. When calculating the thermal resistance of the oxide layer, the property models of solid magnetite and solid spinel are used:

Thermal conductivity of solid magnetite:

$$\lambda_{mag} = 4.23 - 1.37 \times 10^{-3}T \quad (31)$$

Thermal conductivity of solid spinel:

$$\lambda_{sp} = 50.78T^{-0.36} \quad (32)$$

3. Coupling process

When calculating the process of oxidation layer accumulation and corrosion on the surface of the structural material, the time step is usually long, such as about a few hours. It takes hundreds or thousands of hours for the oxide layers to accumulate to a certain considerable thickness. However, the thermal-hydraulic characteristics changes much faster in the system. When calculating the thermal-hydraulic characteristics over time, the time step is usually within seconds. Therefore, the coupling module of thermal-hydraulic and oxidation corrosion typically only calculate the time-dependent changes in reactor thermal-hydraulic characteristics under the influence of oxide layers during long-term stable reactor operations. The final result of this coupling module is shown as Fig. 4.

The thickness of the oxide layer changes much slower than the thermal-hydraulic characteristics of the reactor system, therefore when calculating the thermal-hydraulic characteristics of the system, the thickness of oxide layers, concentration of oxygen and iron and their derivative with respect to time could be assumed to be zero. Therefore, in the coupling calculation process, asynchronous calculation of thermal-hydraulic and oxidation-corrosion modules can be realized. Coupling calculation can be realized by transferring flow field and oxide layer distribution between two modules. The detailed coupling calculation procedure is shown as Fig. 5.

Since the oxidation-corrosion module requires flow field data for calculation, the thermal-hydraulic calculation module would start computing first to obtain a set of stable flow field data. Gear algorithm is used when calculating transient flow field data. When the flow field tends to be stable, thermal-hydraulic calculation paused the values of flow field data and their derivative values will be saved for the thermal-hydraulic calculation in the next loop. Then, the flow field data will be transferred to oxidation calculation module, including temperature and mass flow of each calculation node. When oxidation-corrosion module received the flow field data, the calculation of the time-dependent variation of concentration of oxygen and iron, and the accumulation and the corrosion rates of the oxide layer are conducted.

When the calculation process of oxidation-corrosion stops, the thickness data of oxide layer will be transferred to the thermal-hydraulic module to calculate additional thermal resistance caused by oxide layer, then according to this thickness data, wall heat transfer coefficients will be corrected when the next loop of thermal-hydraulic calculation begins.

4. Calculation objects and setting

4.1. LESMOR reactor

In the present paper, a long-term calculation is conducted for the influence of oxidation-corrosion on thermal-hydraulic characteristics of LESMOR reactor. LESMOR reactor is a small lead-based modular reactor

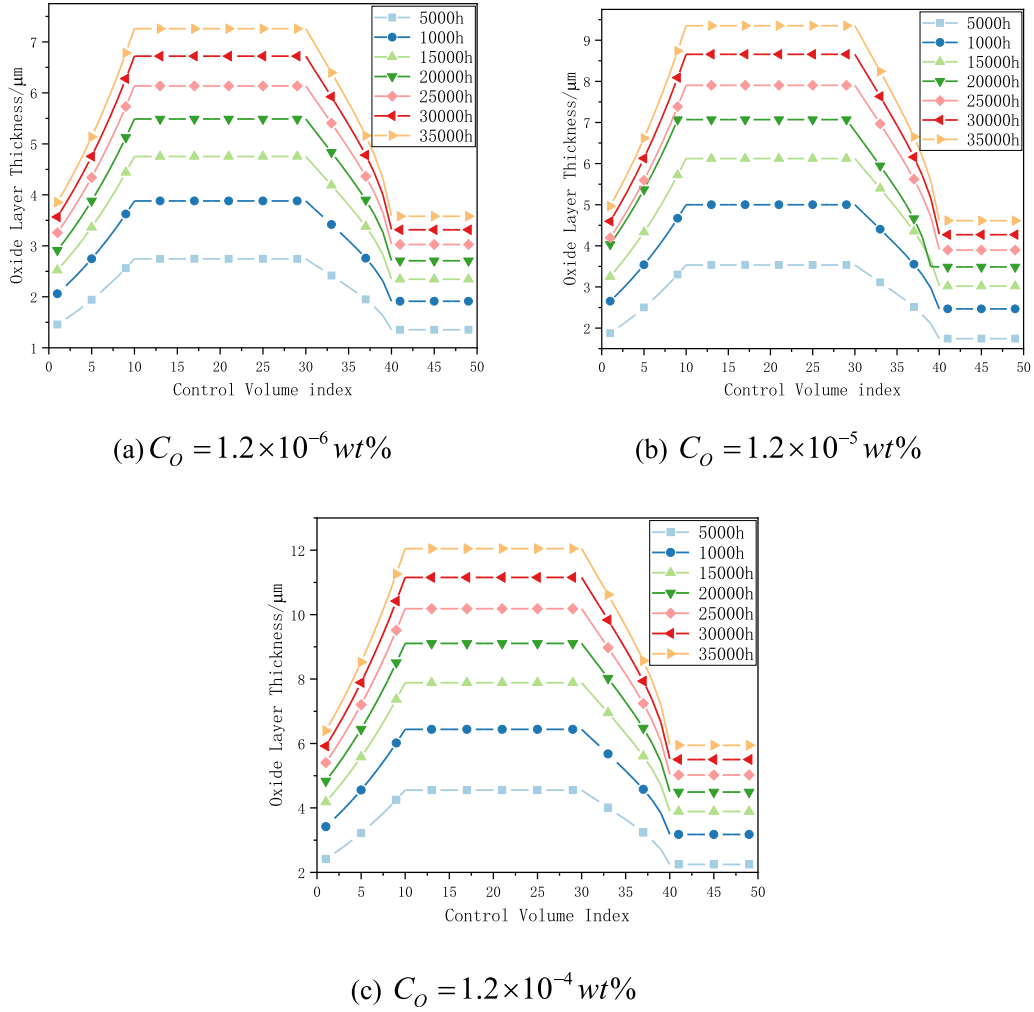


Fig. 8b. Variation of oxide layer thickness with time.

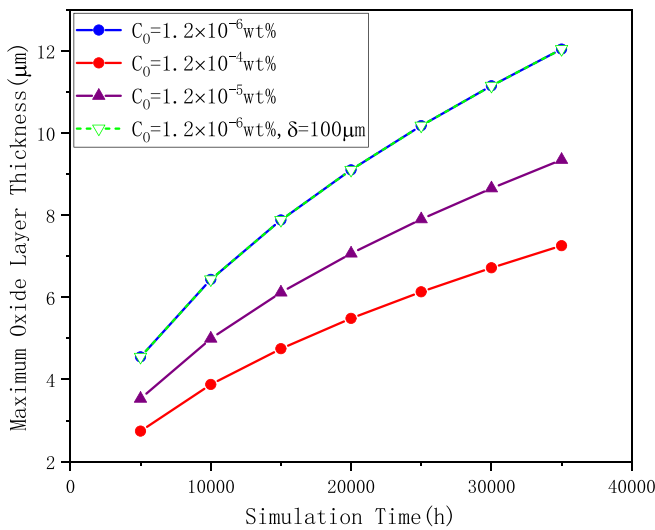


Fig. 9. Variation of maximum oxide layer.

designed by XJTU(Wei et al., 2021). The LESMOR reactor primary circuit adopts the integrated pool arrangement to eliminate the coolant pipeline and reduce the probability of coolant pipeline rupture. The primary circuit is provided with a driving pump, which is started or

closed according to the operation mode. The second circuit system adopts a steam system with a pressure of 3 MPa. In addition, to ensure the safe and reliable operation of the reactor, LESMOR is also equipped with an air-cooled residual heat removal system based on the main heat exchanger. The overall system layout is shown in Fig. 6(a). The primary loop of LESMOR is all included in the reactor vessel, including reactor core, hot pool, heat exchanger and cold pool. The upriser and down-comer are divided using basket. The schematic of LESMOR reactor core is shown as Fig. 6(b).

Fig. 6(a) System layout of LESMOR reactor.

Fig. 6(b) Schematic of LESMOR reactor core.

The detailed structure of the reactor is not considered in this paper. The distribution of the oxide layer throughout the system and the temperature through the whole system is considered. The oxygen control box is emplaced on the riser of the reactor system. The nodalized model of the reactor is shown as Fig. 7.

4.2. Simulation settings

The main simulation parameters are shown in Table 1. In this paper, the influence of initial oxygen concentration and roughness of pipe wall on the thermal hydraulic characteristics of the reactor is conducted. Thus, multiple sets of initial oxygen concentration and roughness of wall are set. The initial oxygen concentration is set from $1.2 \times 10^{-6} \text{ wt\%}$ to $1.2 \times 10^{-4} \text{ wt\%}$, and the roughness of pipe wall is set from 1 nm to 100 nm. Multiple working condition have been set in the present study.

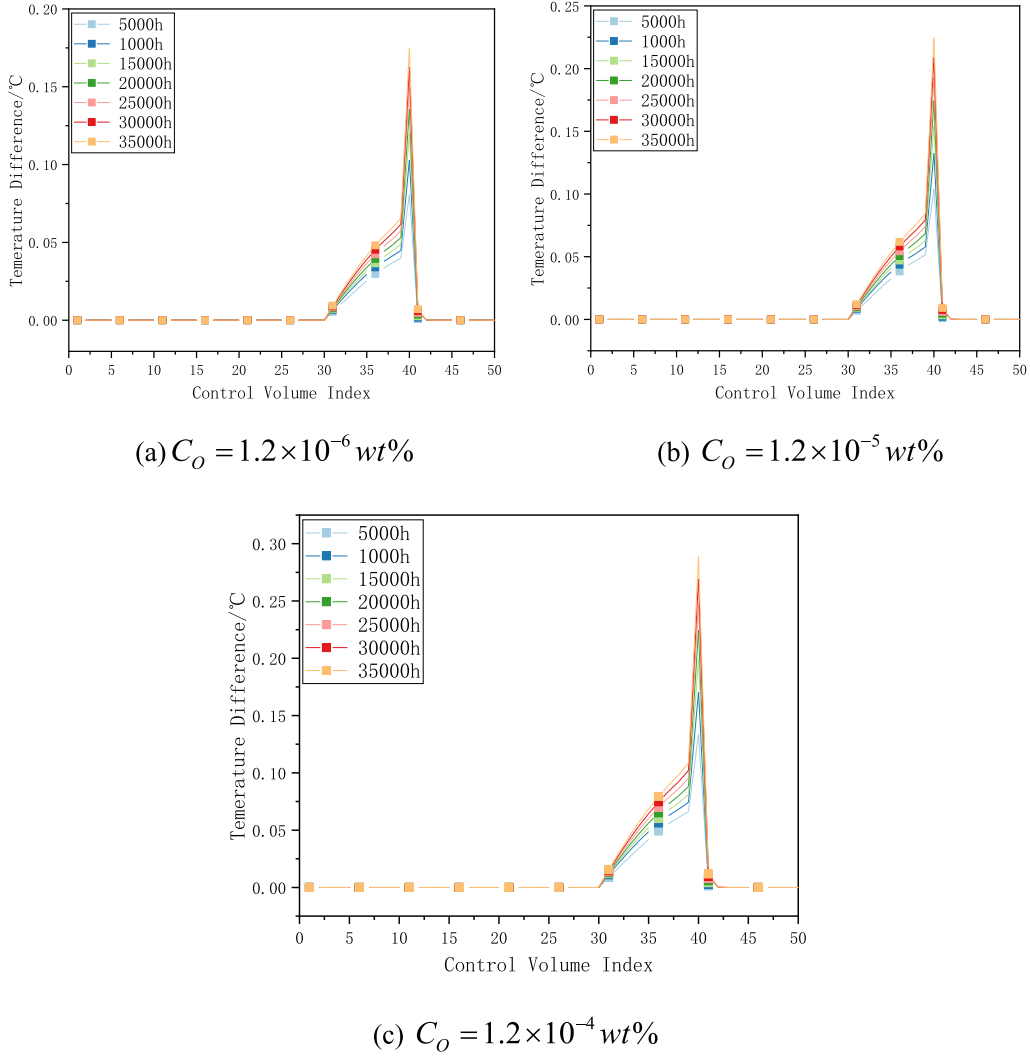


Fig. 10. The distribution of fluid temperature in the reactor system.

5. Simulation result

5.1. Variation of oxide layer thickness over time

Based on LETHAC code, the distribution of oxide layer in the reactor can be obtained. The thickness of oxide layer when initial oxygen concentration in the oxygen control box $C_O = 1.2 \times 10^{-4} \text{ wt\%}$, and the calculation time is 5000 h is shown in Fig 8a. Fig 8b. The maximum thickness of the oxide layer is $11.6 \mu\text{m}$ and the minimum thickness is $2.24 \mu\text{m}$. The maximum thickness of the oxide layer occurs in the riser and the hot pool of the reactor. The minimum thickness occurs in the cold pool of the reactor. This calculation result is reasonable and in line with expectations. After a long period of operation, the oxygen concentration in the reactor remains roughly uniform, and according to Eqs. (7), the growth rate of the oxide layer is positively correlated with the temperature. Thus, the higher the temperature, the higher the thickness of the oxide layer after a long period of operation.

The time-dependent variation of the oxide layer thickness under different initial oxidation concentration is as shown in Fig. 9. The thickness of oxide layer rises rapidly at the first two iterations, then it slows down in the remain iterations. The thickness of oxide layer changes at the heat exchanger and reactor core. The inlet of the heat exchanger and outlet of the reactor core have the highest oxide layer thickness.

Since the higher the temperature, the faster the oxide layer accumulate, oxide layer thickness in the hot pool is the highest while it is lowest in the cold pool. the temperature in the hot pool and cold pool is the highest and lowest respectively.

The variation of maximum oxide layer thickness with time is shown in Fig. 9. The thickness of the oxide layer increase rapidly in the early stage of simulation, and then slows down in the late stage. The overall trend of oxide layer growth shows a parabolic trend which basically proves the rationality of the calculation results.

As can be seen from the figure, the growth curves of the oxide layer almost coincide when the roughness is present and when the roughness is absent, indicating that the existence of roughness does not affect the growth trend of the oxide layer. Theoretically speaking, the existence of roughness will affect the wall heat transfer coefficient of the system, then it affects the fluid temperature, and finally affect the growth rate of the oxide layer. Therefore, the presence of wall roughness would make the growth of the oxide layer more vigorous. However, it can be seen from Eq. (13) that the roughness has little influence on the heat transfer performance, resulting in that the roughness affects less the growth of the oxide layer. The reason is that the effect of roughness on fluid temperature is not significant enough. The present of the roughness could only make 0.1°C impact on fluid temperature, which is too small to effect the accumulation process of the oxide layer. Detailed

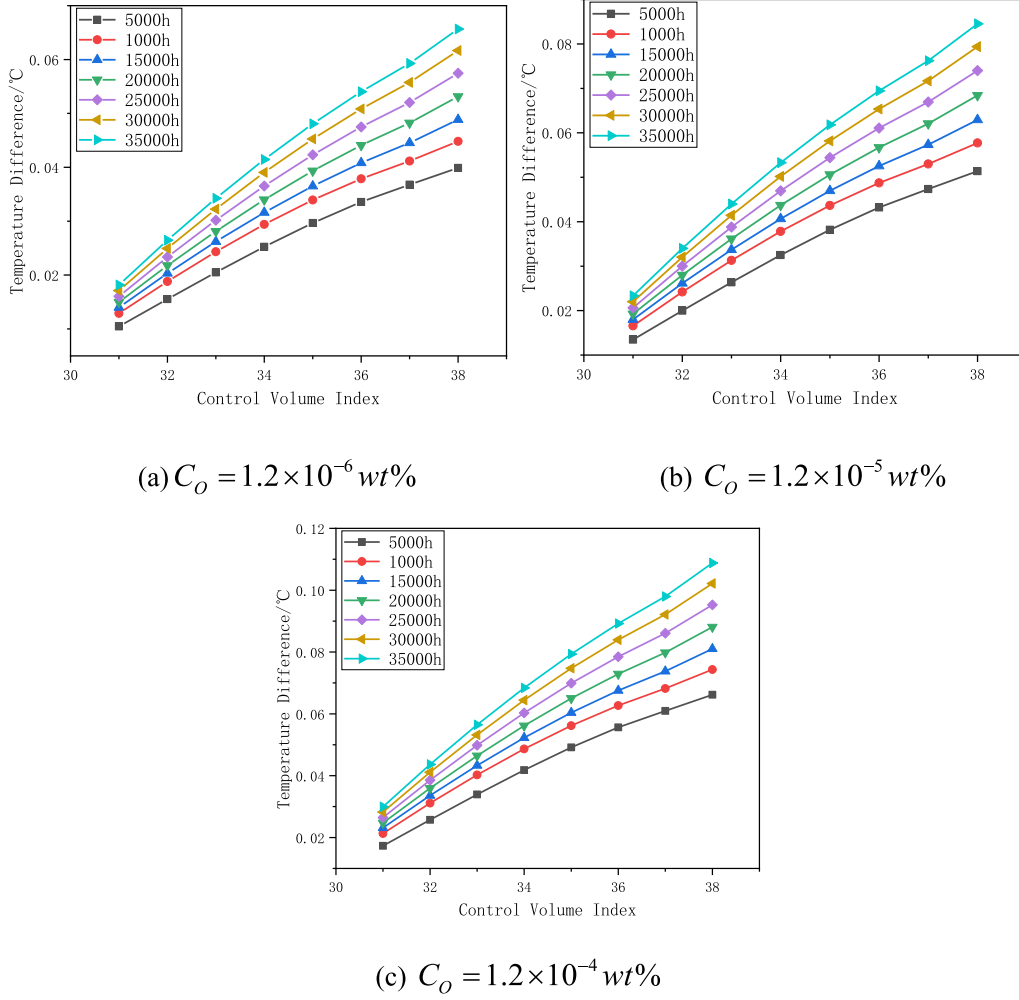


Fig. 11. The distribution of fluid temperature change values in the heat exchanger.

description of the effect of the oxide layer on the reactor fluid temperature can be found in [section 5.4](#).

5.2. Fluid temperature distribution in the system

In order to investigate the change of steady-state thermal hydraulic parameters of the reactor during long-term operation, the long-term change value of temperature is calculated (alias temperature change value), which is defined as the temperature at the current time minus the temperature at the initial time, as shown in Equation (33).

$$T_s(t) = T(t) - T(0) \quad (33)$$

Fig. 10 shows the temperature change value over time when the oxygen concentration is $1.2 \times 10^{-4} \text{ wt\%}$, $1.2 \times 10^{-5} \text{ wt\%}$ and $1.2 \times 10^{-6} \text{ wt\%}$ respectively. As can be seen from the figure, with the passage of time, the fluid temperature is rising, and the heat exchanger section (the 30th to the 40th control volume) has the most obvious rise. The temperature change value is the lowest at the inlet and the highest at the outlet, and it gradually increases from the inlet to the outlet inside the heat exchanger. The maximum temperature change in different oxygen concentration is 0.17°C , 0.22°C , 0.29°C respectively. It is worth mentioning that the temperature deviation quickly disappears after leaving the heat exchanger and before entering the core, which may be due to the flow instability during the calculation of the program.

Since the temperature change value relative from the initial moment of time reaches the highest in the heat exchanger section, the

temperature change values in the heat exchanger section are plotted in Fig. 11. Temperature change values are roughly linearly distributed in the heat exchanger section, and reach the highest value at the outlet. Since the heat resistance of the heat exchanger rises due to the accumulation of the oxide layer, heat in the fluid of the primary loop would be harder to be removed. thus, the fluid temperature of the primary loop rises with the accumulation of the oxide layer.

5.3. Wall temperature distribution in the system

Fig. 12 shows the distribution of wall temperature changes in the reactor system. Here, the wall temperature is actually the temperature of the metal wall below the oxide layer. As can be seen from the figure, with the increase of time, the wall temperature in the reactor hot pool and descent section (corresponding control volume index is between 10 and 30) has a slight rise, but the wall temperature in the heat exchanger section (corresponding control volume index is between 30 and 40) has a certain decrease. When calculating the steady-state characteristics of the reactor, the flow rate and inlet temperature at the secondary side remain constant. This means that the heat transfer capacity of the secondary side will also remain the same during the whole calculation procedure. The maximum temperature change values are -1.7°C , -2.4°C and -3.3°C .

The temperature change values in the heat exchanger are also plotted in Fig. 13. The temperature change value changes slow at the inlet of the heat exchanger, then change rapidly at the outlet. The temperature

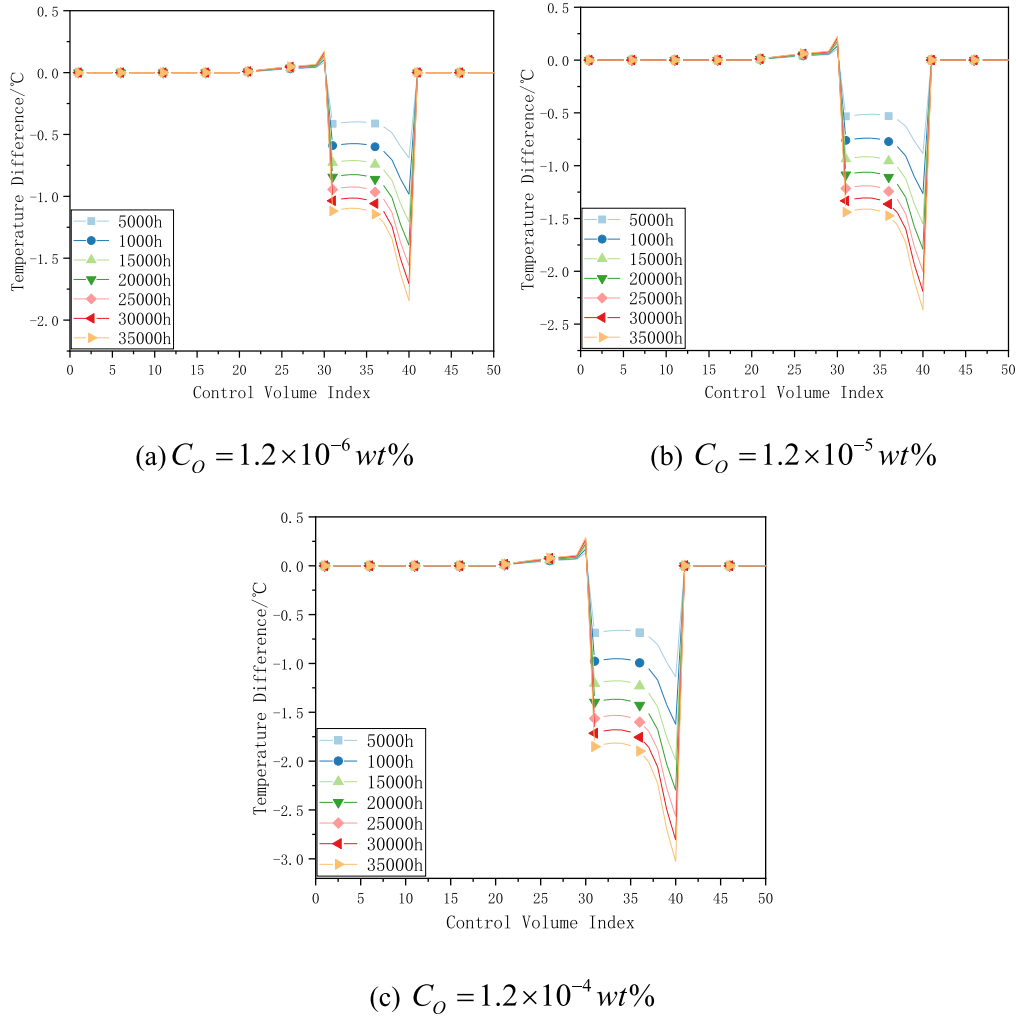


Fig. 12. The distribution of wall temperature in the reactor system.

change value at the outlet of the heat exchanger section reaches the lowest, which are -1.9°C , -2.4°C and -3.2°C respectively.

When simulating the reactor system, the secondary side is to have constant inlet enthalpy and flow, which means that the energy output from the secondary side is constant. When oxide layer accumulates, the heat resistance of the heat exchanger also increases. Because the material thermal conductivity of the oxide layer is lower than that of the steel material, the thermal conductivity resistance increases when the oxide layer accumulates, and the heat transfer of the heat exchanger is hindered. On the one hand, the temperature at the hot fluid side wall increases. On the other hand, the temperature at the cold fluid side wall decreases, and the overall heat transfer efficiency of the heat exchanger decreases.

5.4. Effect of oxidation and corrosion on system temperature

5.4.1. Effect on fluid temperature

Fig. 14(a) show the maximum temperature changes with different oxygen concentration, and Fig. 14(b) shows the maximum temperature changes with different roughness on the surface of the pipe wall. In Fig. 14(a), the maximum temperature change rises as the time goes by. The growth trend is roughly linear. This is because on a larger time scale, the increasing trend of oxide layer thickness is roughly linear. Therefore, the temperature change brought by the oxide layer is liner. The maximum temperature change at the end of the simulation is $0.12^\circ\text{C} \sim 0.25^\circ\text{C}$, and when $C_O = 1.2 \times 10^{-4} \text{ wt}\%$, the maximum temperature

change is highest, the maximum temperature change is 0.25°C , and when $C_O = 1.2 \times 10^{-6} \text{ wt}\%$, the maximum temperature change is lowest, which is 0.12°C .

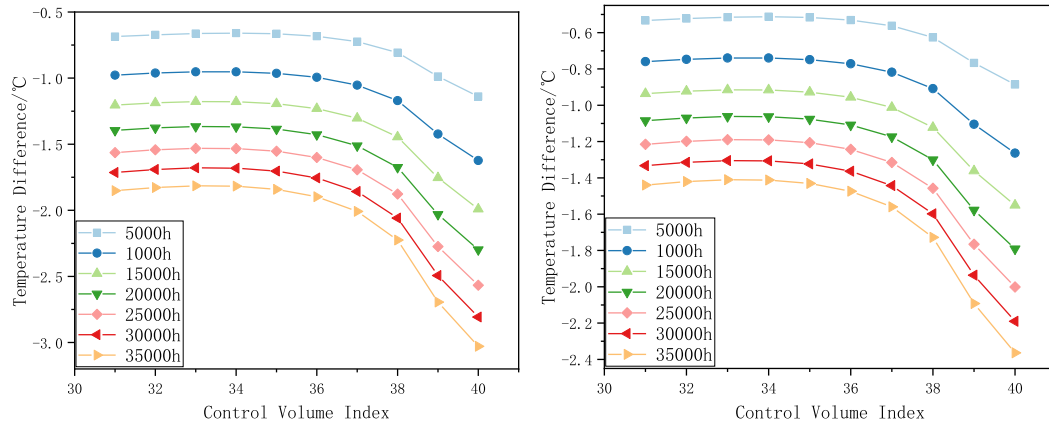
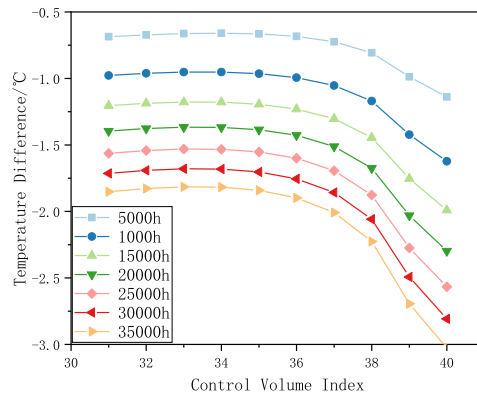
In the Fig. 14(b), the effect of the wall roughness on the temperature change is shown.

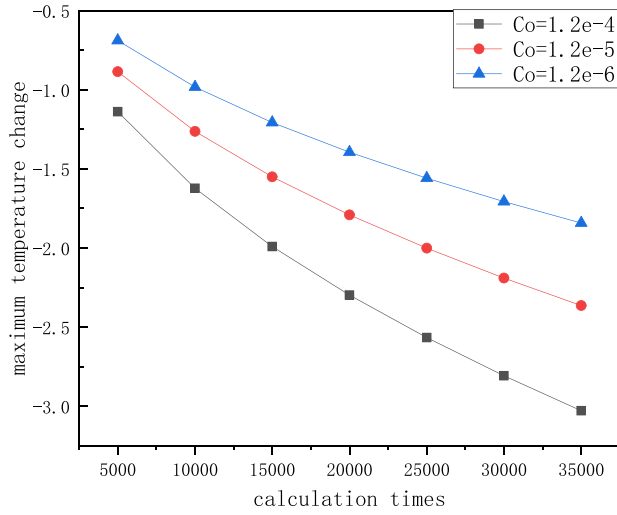
The effect of roughness in Eqs. (15) can be derived. The Nusselt number could be changed by the factor $(\frac{\delta}{D})^{0.48967}$. When roughness increases, the heat transfer capacity in the heat exchanger rises, the heat resistance in the primary side of the heat exchanger declines with the rise of the roughness. Thus, the effect of the present of the oxide layer is weakened. The maximum temperature change values would go down with the rise of the roughness.

5.4.2. Effect on wall temperature

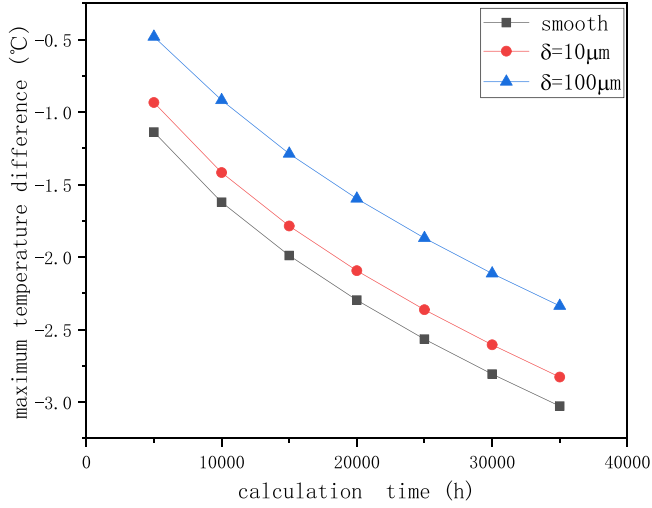
Fig. 15 Shows the effect of the presence of the oxide layer on wall temperature. Fig. 15(a) shows the maximum temperature change value under different oxygen concentration when the wall has no roughness. Fig. 15(b) shows the maximum temperature change value under different wall roughness when the oxygen concentration is wt%. In Fig. 15(a), the higher the oxygen concentration, the greater effect on the wall temperature. The maximum temperature change can be -3.0°C when the oxygen concentration is wt%.

Since wall roughness could affect the heat transfers capacity by affecting Nusselt number, and the present of roughness could decline the heat resistance of the heat exchanger, the higher the roughness is, the smaller effect on wall temperature is. When roughness is $100 \mu\text{m}$,

(a) $C_O = 1.2 \times 10^{-6} \text{ wt\%}$ (b) $C_O = 1.2 \times 10^{-5} \text{ wt\%}$ (c) $C_O = 1.2 \times 10^{-4} \text{ wt\%}$ **Fig. 13.** The distribution of wall temperature in the heat exchanger.



(a) With different oxide concentration



(b) With different surface roughness

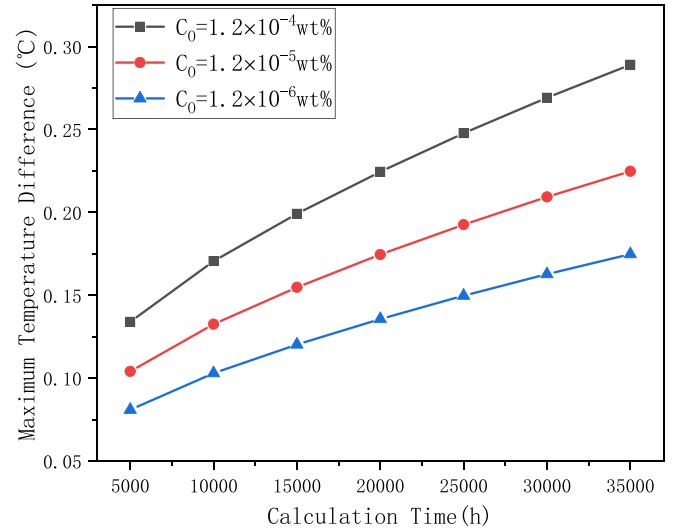
Fig. 14. Maximum wall temperature changes with time.

maximum wall temperature change value could be -2.3°C , while it is -3.0°C when the surface of wall is smooth.

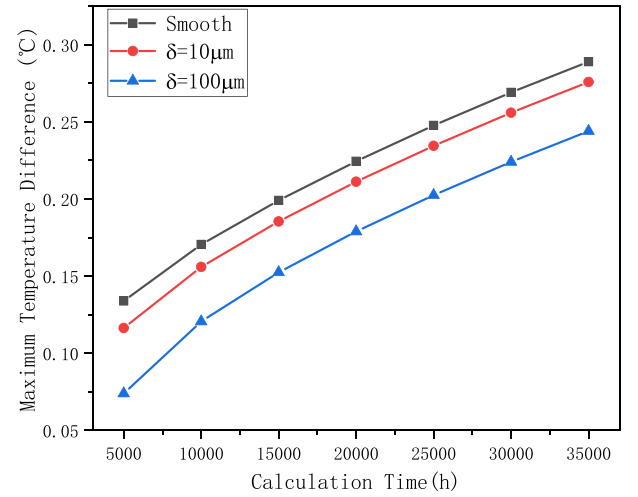
6. Conclusion

In the present paper, a new model aims to investigate the long-term effect of the oxidation and corrosion on the thermal-hydraulic characteristics of LBE reactor is built. To develop and implement this model, the basic behavior and governing equation of iron and oxygen in the LBE fluid is conducted. The model of accumulation and corrosion of oxide layer is developed. Using experimental method and theoretical method, the heat transfer correction model of oxide layer is conducted. Then, based on LETHAC code, the coupling module of thermal hydraulic calculation and oxidation corrosion module is developed and tested. To test and evaluate the reliability of models and calculation codes, a set of simulation of long-term steady state operation of LESMOR reactor was carried out under different oxygen concentration of 1.2×10^{-6} wt%, 1.2×10^{-5} wt% and 1.2×10^{-4} wt%. The detailed conclusion is as follows:

1. With time goes by, the overall thickness of oxide layer increases with time in a parabolic trend. The higher the oxygen concentration is, the



(a) With different oxide concentration



(b) With different surface roughness

Fig. 15. Maximum fluid temperature change value with time.

higher the final oxide layer thickness is. The maximum oxide layer thickness at different oxygen concentration is respectively $6.5 \mu\text{m}$ for 1.2×10^{-6} wt%, $7.2 \mu\text{m}$ 1.2×10^{-5} wt% and $9.0 \mu\text{m}$ 1.2×10^{-4} wt% when the simulation time is 35000 h.

2. The presence of oxide layer will affect the fluid temperature and wall temperature of the reactor. The longer the run time is, the greater the impact is. The change of fluid temperature and wall temperature are obvious in the heat exchanger. The maximum temperature change could be 0.29°C for the fluid, and -3.0°C for the wall.
3. The greater the initial oxygen concentration is, the greater the thickness of the oxide layer is, and also the greater the influence on the thermal hydraulic characteristics of the reactor is.
4. The presence of the pipe wall roughness can weaken the effect of the oxygen layer. When the roughness is $100 \mu\text{m}$ and oxygen concentration is 1.2×10^{-4} wt%, the maximum fluid temperature change value could decline 0.04°C , and the maximum wall temperature change value could decline 1.4°C when the simulation time is 35000 h.

Since the operation temperature of LESMOR reactor is relatively low, the oxide layer thickness after long-term operation is lower than the reactor with higher operation temperature. Therefore, the effect on the thermal-hydraulic characteristics, especially fluid temperature and wall temperature, would be less significant. When reactor operation temperature is higher, such as 600–800 °C, the fluid temperature and wall temperature of the reactor would change much more obviously, which will affect the safety of the reactor after long-term operation. The impact of the oxide layer formed by the oxidation corrosion of lead bismuth on the operation of high-temperature and high-speed reactors cannot be ignored, especially in the analysis of the long-term safe operation characteristics of reactors. As mentioned in the article, the temperature changes in the reactor core and the impact of heat exchange efficiency of heat exchangers are the focus of reactor safety analysis. In the future, we will also conduct oxide layer accumulation on higher operating temperatures. Analysis of the impact of the reactor. More complex and significant impacts will soon appear in the results.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

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